Engineering Mechanics 1 · Unit_02 · UNIT 2

DRAFT — UE26ME142A · Unit_02 · UNIT 2 · UE26ME142A Unit 2 Class 22 AV Summary

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

UE26ME142A Unit 2 Class 22 AV Summary

1 UE26ME142A Unit 2 Class 22 AV Summary

Study focus: UE26ME142A_Unit 2_Class 22_AV Summary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000202 · EV0000203

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr Svinkitesh, a civil engineer from the Department of Science and Humanities, PS University.

Key details

- in the last class we had discussed about unit two equilibrium chapter and also we had discussed few problems on equilibrium chapter.
- Today we will be starting with the next chapter, that is chapter number four.
- from the textbook which is on trusses under unit two as prescribed by the university PS University.
- A truss is a structural member which actually carries the load from the roofing element or a slab or a deck increase of a bridge.
- And transfers the load coming on this roofing elements to the column on either side and finally this load will get transferred to the foundation.
- From foundation it will get transferred to the ground below.
- The stringers are supported by this rafters or cross beams as they are called.
- And this cross beams are attached to this structure, which you can see on either side of the bridge.
- There is one here and there's one more here.
- Which are nothing but the actual truss.
- The load from the vehicle and the Dex Lab will transfer to this stranger.
- Then the string from stranger it will get transferred to the cross vehicle.
- Wherever there are joints.
- Here also you will have a load L.

- to each other and you will have the vehicle moving here and in case if it is a roof truss, usually the roof trusses are going to be slope.
- Then from the purlin or the cross staff, it will get transferred to the trusses on either side.
- A truss is basically used when the span between the supports.
- There will be another half on the left side of the screen.
- And you will have a support on either the side.
- Then we go for such type of structure.
- And also when we are not intended to construct an upper floor above the truss, we go for truss.
- And usually the truss will be constructed of either wood.
- Earlier we used to do it in wood, but nowadays since it is very difficult to get wood and also there is restriction from the government for the use of wood.
- we have moved on to using iron or steel.
- It can also be manufactured using aluminium or any other suitable metal okay for the purpose.

Example or application from the lecture

- This is the Dex Lab and you can see this, they have given it as strangers, okay.
- From cross beam it will get transferred to the truss on either side, and you can see they have given the load at the joint, which is given as L.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000202 · EV0000203 · EV0000204 · +1 source units in pack.json

Core explanation · 05:00–09:59

Constructed of a lighter material, lighter weight material.

Key details

- Vehicles that will be moving on this bridge.
- That weight will be quite large when compared to the self weight of this truss itself.
- The in the problem if it is mentioned to be considered the sulphate, we have to take the sulphate into account.
- Bridge trusses that are employed.
- This case you have a central member and of course it is symmetrical when you see the geometry of the truss, but you can see the inclined members which are moving from top to bottom.
- from left to right.
- And you have four members here.
- Members are the structure which are in between two joints.
- Likewise you have several member here.
- Ends of the member will be will be connected to what is called as a joint.
- Unless you understand a joint, you cannot understand member.
- A member is a structure which is in between two joints.
- But this is a joint where you have two members meeting.

- Plus, of course, you have the support reactions also here.
- This is also a joint where you have three members meeting.
- If you see the left side, the inclined members are moving from left to right, okay, and it is upwards, it's moving upwards.
- Whereas here, except the end, the other was moving from right to left, from up to down, okay.
- And the right side of the members, the income members are moving from top to bottom from right to left.
- Each member moving in a zigzag way like this, the inclined member.
- One member if it is from left to right, the other member is from right to left, top to bottom, and this is from bottom to top.
- And this is a K member you can see that this looks like a K joint, okay?
- Here also it is inverted K joint.
- That is named after the person who has invented this.
- It is based on how you connect the members and how do you take the joints into account.
- One more thing that you should remember in case of processes.
- The members in the trusses will be loaded at the joints only.
- None of the member will be loaded in between.
- That is because.
- The members of a truss can take either a compressive load or a tensile load only.
- Next, coming to what are the different types of roof trusses that you have, there are four different types that is mentioned here.

Example or application from the lecture

- For example, if you take the weight of the.
- With this background, let's now look at what are the different types of.
- So the first five which is given here are the trusses for bridges.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000206 · EV0000207 · EV0000208 · +1 source units in pack.json

Core explanation · 10:00–14:59

One is Fink, then it is Pratt.

Key details

- Again Hovi and Warner.
- Here you have.
- Member from going from the incline members going from bottom to top, from left to right, whereas in this case it is moving from top to bottom from left to right, the left side.
- Whereas the right side, you can see here it is the incline members are moving from.
- Top to bottom from left to right, whereas here it is bottom to top from left to right.
- And of course they are symmetrical, all of them are symmetrical here.
- Now you have to remember these type of trusses because in the examination, ISA or ESA they can ask you to draw a.

- A Pratt trust or a hobby trust, or they can give you a trust and ask you to name the type of trust.
- A plane truss is a truss in which you have all the members and the joints in the same plane.
- If you take it, it is in the horizontal plane only.
- As I have told you that the members of a truss is made up of either only compression or only tension.
- Tension and compression will have a pair of forces, either they will be moving away from each other or moving towards.
- If the external forces are moving away from each other, the body would tend to elongate.
- Here it will be moves towards this joint.
- Here it will be moving towards this joint.
- Such forces are defined as tensile force.
- Comping coming to compressive force.
- If the external forces are moving towards each other along the longitudinal direction of the member, the member will try to get compressed.
- Such members are called as compressive members.
- The bottom half and take it separately, you can see here.
- This is external force, compressive force, C, the internal force will be acting towards the joint for this half of it.
- Whereas if you take the upper half, the external the internal force will be moving towards this.
- The internal forces that is away from each other, the external forces will be moving towards each other, the body will be under compression.
- In fact, I am going to repeat this again, taking the tension into consideration.
- That is what is called as tensor member.
- Now, because of the external forces, if I take the internal forces, the internal forces will be acting towards each other, that is towards the joint.
- The body would get elongated, so it is under tension.

Example or application from the lecture

- For example, if you take here, you have three members here, all these three members.
- So here is an example of what is tension and what is compression.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000210 · EV0000211 · EV0000212 · +1 source units in pack.json

Core explanation · 15:00–19:59

Now, what because of what these forces are coming?

Key details

- Because of the external force that is acting at the joint.
- And as I told you, in case of trusses, the lore cannot be applied in between the members.
- It has to be applied at the joint only.
- These are external forces.

- Because of these external forces, you get the forces in the members which are internal.
- This one also is under compression, whereas this one is moving away from the joint.
- Kindly remember this because this is going to be helpful when we are going to be solving problems on trusses.
- And especially in case of trusses.
- What we are going to be doing is to determine what is what are the forces that is acting in each of the member and what is the type of force that is acting, whether it is compressive or tensile.
- And how do we determine the forces in the members?
- There are two methods.
- First method is method of joints, the second method is method of section.
- Now your syllabus you have only method of joints and that is what we are going to be concentrating on.
- The end A is supported with a hinge support or a pin support, so you have two reactions, one vertical and one horizontal, and end B is supported with a roller support, so you have one reaction which will be acting vertically upwards.
- We have to first determine the support reactions.
- In this case, to identify a joint which will have the number of unknowns or the number of members where the unknowns are to be determined, which are the member forces, which are only two members are there.
- So in case of method of joints, what we do is we take each joint separately as indicated here.
- You can see here they have indicated the joint A here.
- No other forces are acting at any of the joint.
- The reactions are also going to be vertical.
- And they have written the diagram here.
- Considering the joint A only separately here.
- This is joint A where you have a reaction at A acting which is written as R.
- Then you have the member AF as well as the member AB.
- The member EAF, it is considered to be under compression.
- That means forces will be acting towards the joint.
- You can see that is what is considered here.
- See here the member internal forces is acting towards the external forces will be acting away.
- Whereas the member A B it is considered to be in tension, so that means the forces will be acting away from the joint.
- This is the.
- Force that is reacting external force.
- That is algebraic sum of all the forces along X to be equal to 0 and algebraic force of all the forces along Y to be equal to 0.
- I can determine the forces in the member AF and AB.
- Once I find the forces in these two members, I can next go to the next joint, which will have only two unknowns.
- That is, the next joint is A is F after A.
- I am taking the next joint as F.
- The m force in the memory AF is already determined from this joint.

- The sense of the force in this is moving towards the joint again.

Example or application from the lecture

- Now there's a simple example that is given here, a roof thrust.
- So we have an example here which is a roofing truss that is taken.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000214 · EV0000215 · EV0000216 · +1 source units in pack.json

Core explanation · 20:00–24:59

This one AB will not come into picture when I take the joint Ef, you can see here this is F.

Key details

- Then we will analyze this again by using the equations of equilibrium to find out the these two unknowns, okay, which we can determine by using sigma fx equals 0 and f y equals 0.
- Once I find this, so now we have found out these three members along with this.
- The external force that is applied is also known.
- Then you can take the next joint as.
- Joint C here.
- This member is known.
- These two I can find out.
- The mem force in this member will be zero.
- We will look at that a little bit later.
- Then I can take the force in the in the joint E.
- We have already determined the force in this, that is zero member in this.
- Then coming to this joint D, here we would have determined these two earlier itself, and also R2, but we can take R2 as an unknown.
- from these two forces whatever we know, if I can find out R two and I can compare what is the value that I have got earlier to what I will get here.
- See you in the next.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000218 · EV0000219 · EV0000220 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE26ME142A_Unit 2_Class 22_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/022_UE26ME142A_Unit_2_Class_22_AV_Summary__ebcc1256b1d0786a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 22_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/022_UE26ME142A_Unit_2_Class_22_AV_Summary__ebcc1256b1d0786a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 22_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/022_UE26ME142A_Unit_2_Class_22_AV_Summary__ebcc1256b1d0786a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 22_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/022_UE26ME142A_Unit_2_Class_22_AV_Summary__ebcc1256b1d0786a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md have no readable `.slides.md` text; their original page images are included.
2. 4 rendered pages in 015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.slides.md have no readable `.slides.md` text; their original page images are included.

4. 4 rendered pages in 017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.slides.md have no readable `.slides.md` text; their original page images are included.
7. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.slides.md have no readable `.slides.md` text; their original page images are included.
9. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.slides.md have no readable `.slides.md` text; their original page images are included.
10. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.slides.md have no readable `.slides.md` text; their original page images are included.
12. 3 rendered pages in 021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.slides.md have no readable `.slides.md` text; their original page images are included.
14. 3 rendered pages in 023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.slides.md have no readable `.slides.md` text; their original page images are included.
16. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
17. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.

18. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
19. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
20. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00050	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00051	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...
SRC00052	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00053	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...

C Revision and retrieval practice